MODULE PRESENTATION · CENG 396

STRATA

Module 1: Physical Stabilization & Control

Mechanics, actuation, closed-loop control and safety for the one-axis active stage

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We build safety, not assumptions.

Module 1 builds the part of Strata that physically moves

DELIVERABLE

A bench-tested, safely moving active stage

MODULE 1 TEAM

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The motor loop stays with the mechanical design, so it runs without waiting for the camera or dashboard.

Mechanics

Build both linear motion layers, the physical stops and the three-story flexible building model.

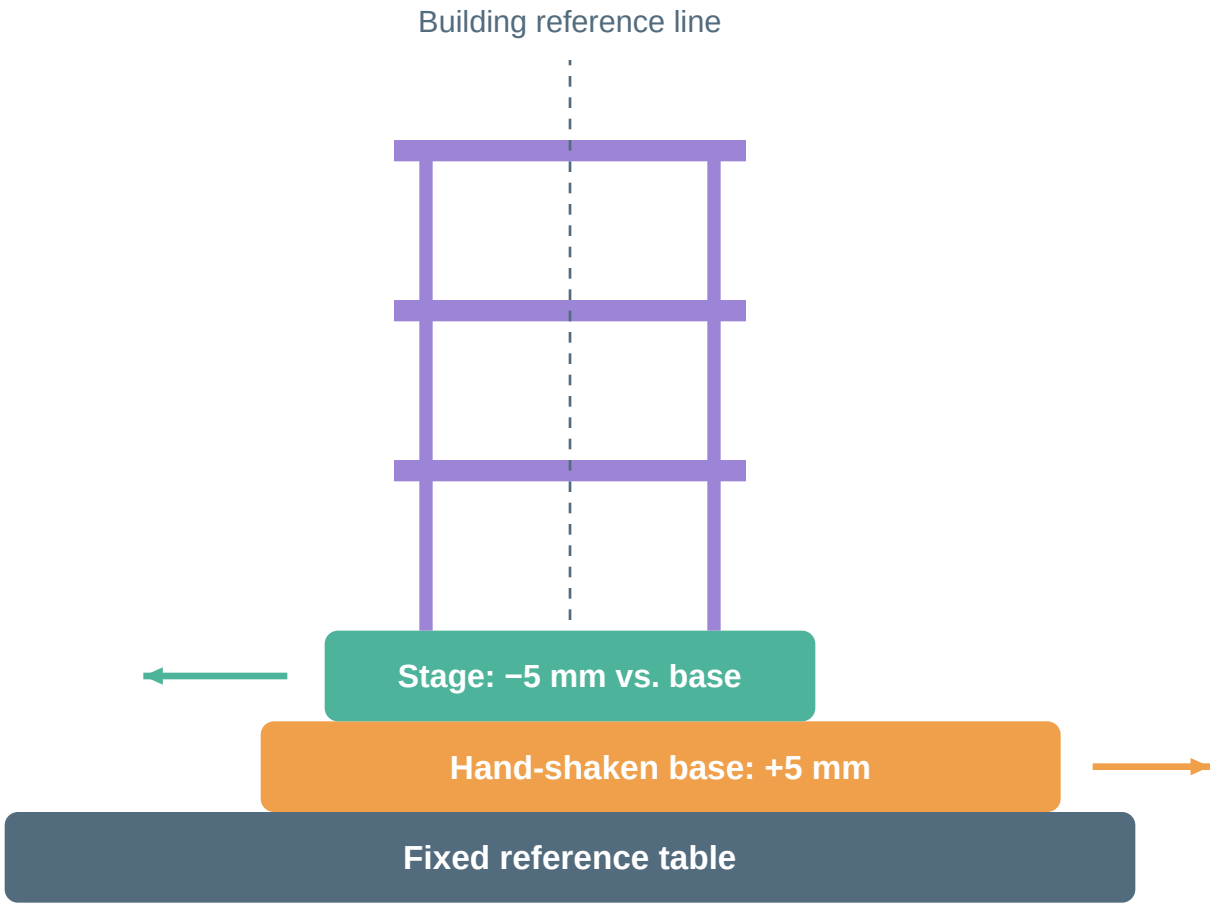
Actuation and sensing

Select the actuator, then measure base vs. table and stage vs. base with encoders or linear sensors.

Control and safety

Implement a closed-loop controller with safe stroke limits, an emergency stop and tested loss-of-signal behavior.

The stage moves against the base so the building stays nearly still



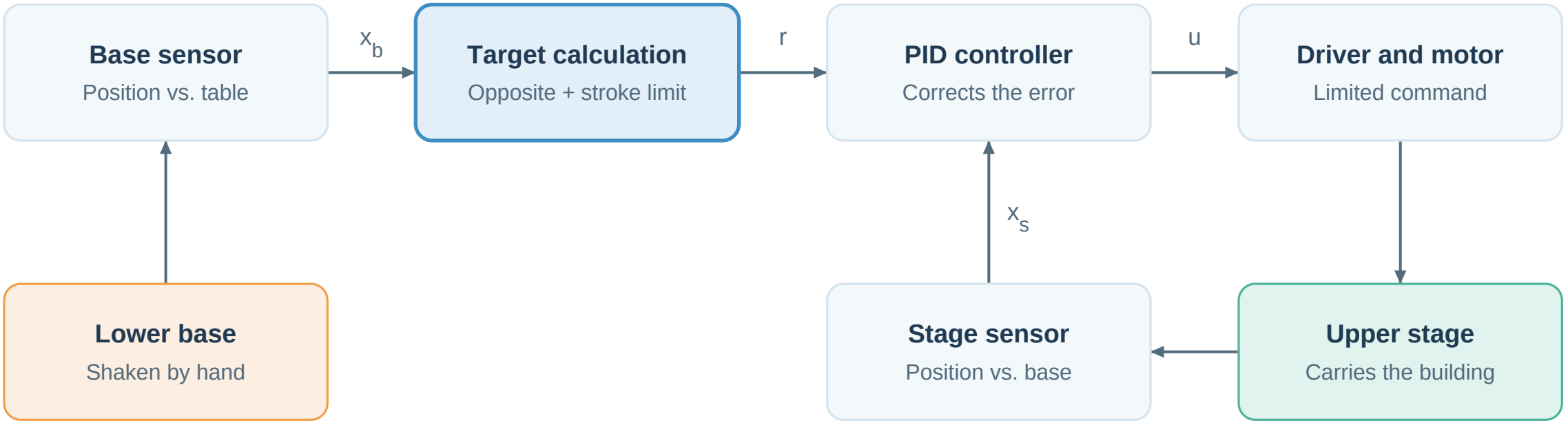
The core equation

Stage vs. table = base vs. table + stage vs. base
The goal is to drive the left side toward zero, so the relative stage motion must mirror the base motion.

Illustrative example	Motion
Base vs. table	+5 mm
Stage vs. base	−5 mm
Stage vs. table (ideal result)	**0 mm**

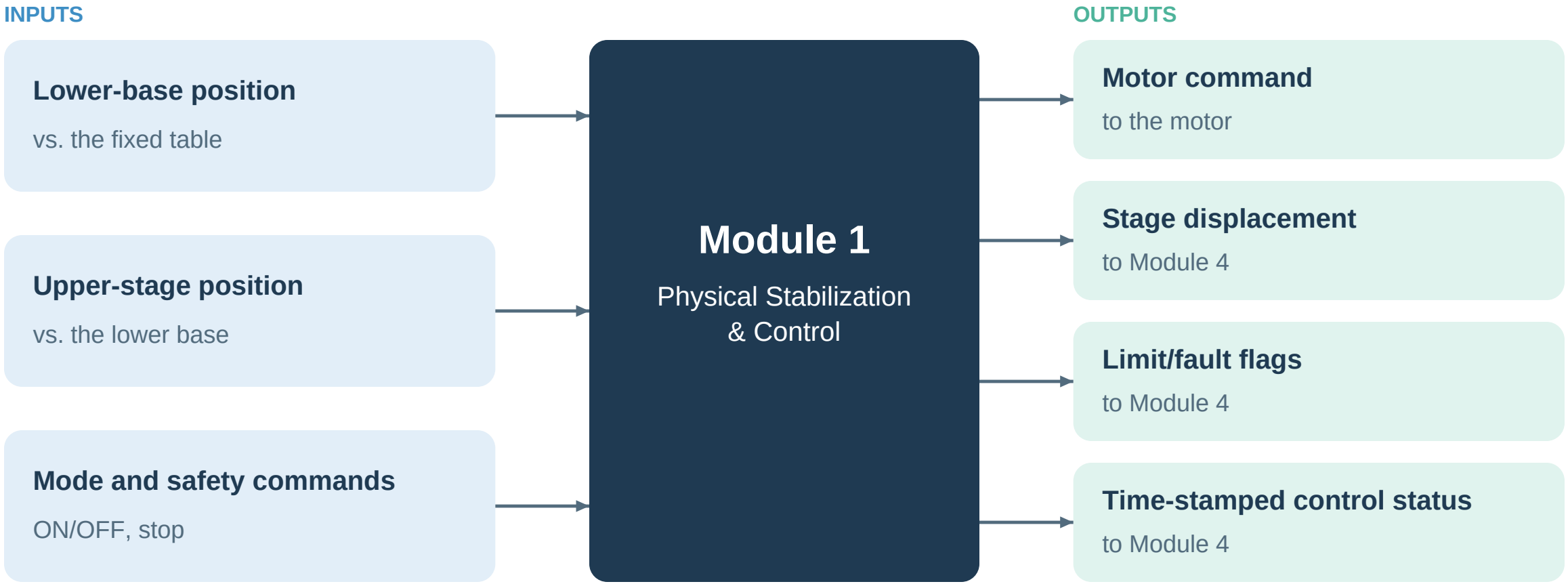
Real performance is limited by stroke, motor capability and delay.

The control loop uses only the two local position sensors



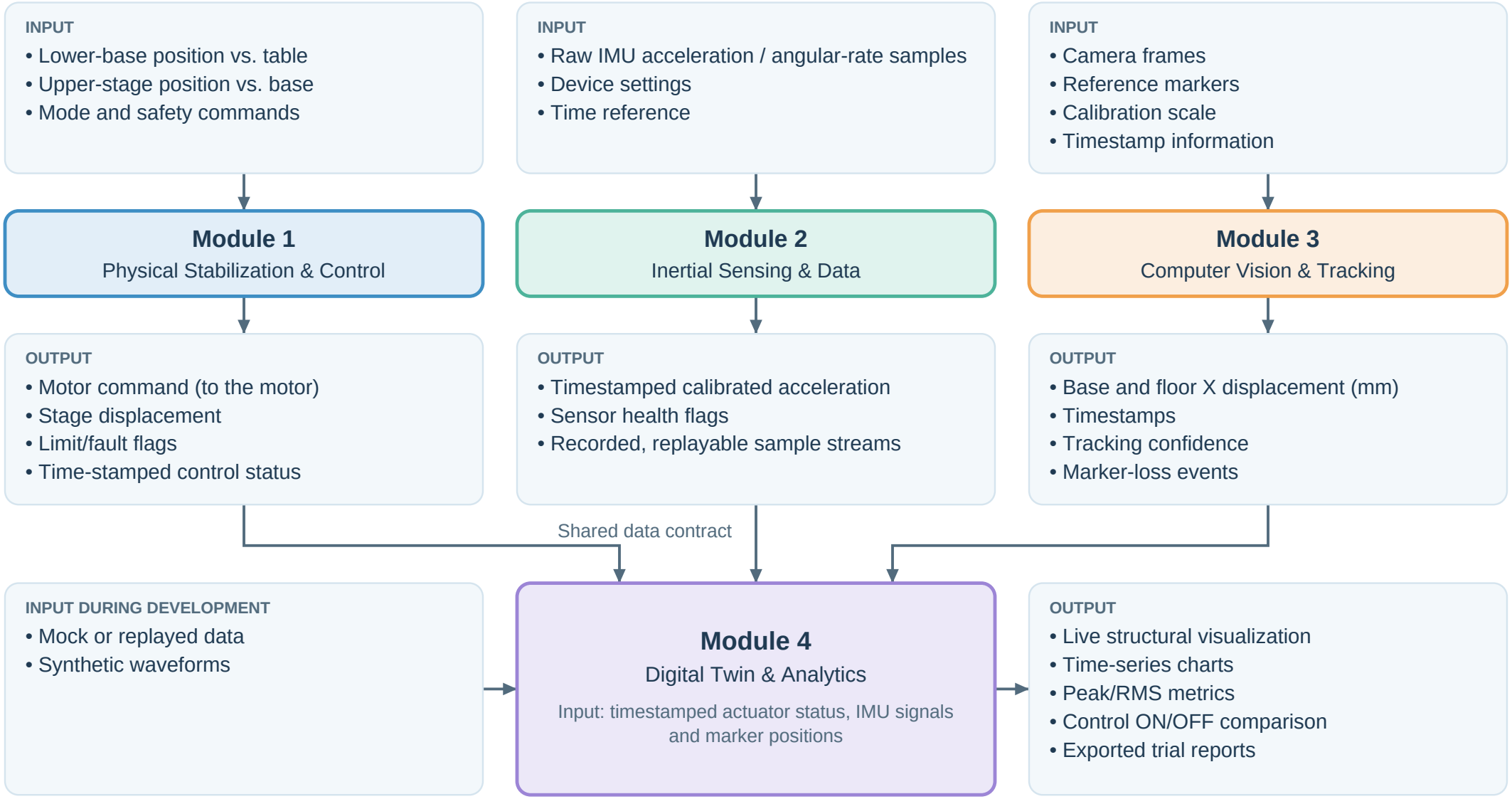
The camera and IMUs observe every trial independently but do not feed the motor loop. The building's response is measured outside this loop.

Module 1 turns two position readings into safe motion



Stand-alone test: follow scripted positions on the bench, then shake the base and measure the stage's absolute motion.

Strata modules: inputs and outputs



Safety is tested before the building goes on

Stroke limits

Software keeps the stage target inside its safe travel.

Physical stops

They limit travel even if the software fails.

Emergency stop

It stops the motion at once, and loss of signal must also be handled safely.

Our test order

1. Scripted positions on the bench
2. Shake the base, measure the stage
3. Test stops and loss of signal
4. Attach the building, compare OFF and ON

Thank You

Questions?



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